3D simulation Oscillator add-on Technical Report

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Contents

0.1 GINGER Setup and Execution 1
0.1.1 How to run GINGER 1
0.2 MATLAB Oscillator Add-On 1
0.2.1 Purpose and workflow of the MATLAB add-on 2
0.2.2 Physical and numerical principles, key equations 3
0.2.3 Test case results 3
0.3 Input Parameters and Configuration Files 3
0.3.1 post processing 5
0.3.2 Description of some parameter selection 5
0.4 Desynchronization Studies 6
0.5 Data Management and File Locations 8
0.6 Version Control and Reproducibility 9
0.7 Future Work or Known Limitations 9

List of Figures

1 Gaussain Beam evolution in cavity 4
2 Cavity schematics 7
3 Radiation Peak Power vs. Pass number 8
4 Radiation Peak position vs. oscillator pass number 8
5 Radiation FWHM 9

List of Tables

Abstract

An oscillator add-on module was developed for FEL (Free electron Laser) `GINGER-3D` package. The add-on module was tested using the oscillator FEL parameters of our FEL lab at the University of Hawaiʻi at Mānoa (UHM). In this technical report we describe the add-on in detail and the instructions for executing.

0.1 GINGER Setup and Execution

In this section we give a description of the `GINGER-3D` files, installations and how to run (only reference to GINGER-H [3] is available) . To the writing of the current report the `GINGER-3D` was run on Mac, but can be run on Linux.
Two main files which are supplied by William Fawley (making of `GINGER-3D`) according to your computer instructions:

- xg3D-static-1.1.0-a5
- xg3DPP-M2MAC-DISLIN-static-1.1.0-a

The first file is the GINGER3D executable, the second file is the post-processor. No installation or compilations is needed but other libraries might be necessary based on your computer platform. The manual is found online, latest version of the manual is only supplied by William.

0.1.1 How to run GINGER

`GINGER-3D` must be run with an "in" (input) file supplied. An output hdf5 file is generated at the end of each run, this file can be opened with the post-processor program to produce a POST script or a PDF file.
Upon request `GINGER-3D` can supply a FLD file, which contains the radiation field at the end of the undulator. The FLD file is a complex time sliced 2D field. A particle distribution, RST file can also be outputted and saved.
The FLD file is used throughout the add-on, while the RST file can be used to create the electron bunch energy-time plot.
We send you to the `GINGER-3D` manual for further investigation.

0.2 MATLAB Oscillator Add-On

The following files are included in the installation directory:

1. *osc_add_on.m* - Main file which is used to run the code.
2. $/ginger/MK3_make_G3d_in_file.m$ - A function for generating the input file for the first oscillator pass.
3. $/ginger/MK3_make_G3d_2in_file.m$ - A function for generating the input files for the second oscillator pass and on.
4. $/ginger/input_EE.m$ - A helper file for running `GINGER-3D` and functions for generating the input files.
5. *desynchronize.m* - Function responsible for desynchronizing the oscillator cavity.
6. *inser_lens2.m* - Function file for the Fourier optics, responsible for Natrual diffraction and mirror reflectivity.

7. *my_funct.m* - Function file which is used to return data for displaying the optical pulse in time and frequency domain (name should be changed).

8. *my_fwhm.m* - Function file which calculates and return the width and peak position of the radiation pulse.

9. *find_waistxy.m* - Function file which finds the transverse sigma size of the beam in the x and y axes.

10. *Zray.m* - This function file returns the ZRayleigh and waist size.

11. */test/intput_tst.m* - This file is used for running a stand alone version of the add-on.

0.2.1 Purpose and workflow of the MATLAB add-on

There is no need to run any external functions; the main file does all the work. It is necessary to update the parameters of your work case scenario, in the beginning of the main file:

- *osc_add_on.m* (1)

To run a test of the add-on with five oscillator passes, set the parameter "TSTIN" to 1 at the configuration part at the beginning of the main file.

The main file is divided into three parts:

1. The first oscillator pass.

2. A "for" loop of the second oscillator pass to the end of the oscillator passes.

3. displaying the results.

Each oscillator pass (part 1,2) is combined of:

- Running `GINGER-3D` (*input_EE* function) getting the time sliced 2D complex radiation field at the end of the undulator (from the "FLD" HF5 file). From the second oscillator pass, the time sliced 2D complex field from the previous run is inserted to `GINGER-3D`, reading the "FLD" file.

- Running through the oscillator cavity.

- Implementing desynchronization and cavity loss

- getting optical pulse in time and frequency for display.

0.2.2 Physical and numerical principles, key equations

The `GINGER-3D` radiation field data consists of a matrix with dimension $(2 * n_t, n_x, n_y)$. This array represents the complex field (real, and imaginary parts) at the end of the undulator with n_t the time slices and n_x, n_y the field values in the transverse plane.

The FEL cavity is solved numerically using Fourier optics.
The complex 3D E-field from `GINGER-3D` output $E_{in} = ar + 1i \cdot ai$ is 2D Fourier transformed $E_f = \mathcal{F}\{E_{in}\}$.
We propagate the field with free-space propagation kernel [1]

$$H_1 = exp(1i \cdot Z_1 \sqrt{k^2 - k_x^2 - k_y^2}) \tag{1}$$

where k is the wave number $k_{x,y}$ is the spatial frequency grid, and Z_1 is the length from the undulator end to the mirror.
The field back in the time domain at the mirror location is: $E_{z1} = \mathcal{F}^{-1}\{E_f * H_1\}$.
At the mirror we use the thin lens transfer function:

$$H_{lens} = exp(-1i \cdot \frac{k}{2f}(X^2 + Y^2) \tag{2}$$

where f is the focal of a lens identical to mirror radius of curvature divided by 2,and X,Y are meshgrids fro spatial coordinates. The field after the mirror plane is $E_{lens} = \mathcal{F}\{E_{z1} * H_{lens}\}$.
Afterwards, we propagate the entire cavity distance Lc to reach the second mirror plane. Transform Fourier again at the mirror plane and propagate another distance to the undulator entrance. A note when choosing parameters, the stability condition of a cavity should be taken into account [7]:

$$0 \leq (1 - \frac{L}{R1})(1 - \frac{L}{R2}) \leq 1 \tag{3}$$

Where L is length of the cavity and R1,2 is the radius of curvature of mirrors.

0.2.3 Test case results

The Oscillator add-on was tested using the UHM parameters with numerous scenarios. The electron bunch duration, the current and desynchronization values were altered (see section on desynchronization). We observed maximal energy extraction when desynchronizaing the cavity length. More energy extraction was observed when the electron bunch was shorter than the slippage length.

Instructions to connect with GINGER output files

0.3 Input Parameters and Configuration Files

Input files are generated automatically, all input parameters should be updated at the start of the main file. Further parameter changing, for example, parabolic bunch distribution instead of gaussian requires changing of the

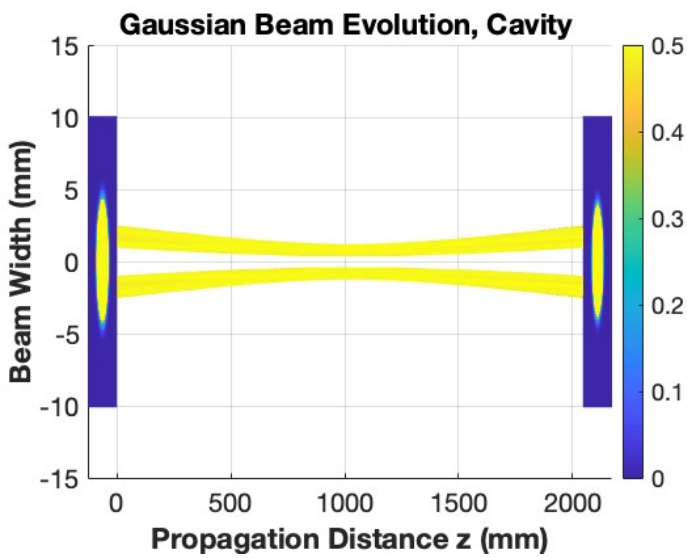

Figure 1: Test case results, gaussian beam evolution in cavity. The cavity starts from z=0 and ends at z=Lc=2.04690m. Beam width is observed in the plot overall 200 roundtrips.The focusing strength f=0.65m

$MK3_make_G3d_in_file.m, MK3_make_G3d_2in_file.m.$

The input parameters for the simulation are:

Beam Parameters:

- Energy [Mev]
- Fractional RMS Energy Spread.
- Normalized Emittance x, y [mm mrad].
- Twiss parameters betax,betay,alphax,alphay.
- Electron bunch Guassain pulse RMS width [s].

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FEL parameters:

- Undulator Period [m].
- RMS Undulator paramter.
- Number of Undulator periods.
- Radiation wavelength [m].

- Undulator length [m]

.
Oscillator parameters:

- Cavity Length [m].
- Mirrors Radius of Curvature [m]
- Mirrors Reflectivity.
- number of passes in the cavity.
- Desynchronization values for simulation.

.
GINGER3D simulation parameters:

- Number of Bunch slices.
- Number of Radiaiton slices.
- Totall number of Photon slices each bunch slice will interact.
- The resolution in x,y.
- Number of radial grid zones.

0.3.1 post processing

In order to generate a movie for the optical pulse in time, "*myvid*" should be set 1. "*SAV*" should be 1 for saving figures. In accordance there is "*sav_dir*", which should be the directory to save, and "nam" which should be equal a given name for figures.

There are numerous files for post-processing of the figures.
If all of the "RST" hdf5 files are saved in a simulation run, by commenting on a part in the *input_ee.m* file, a phase space plot can be generated using *phase_space.m* file.

0.3.2 Description of some parameter selection

The Courant-Snyder or Twiss parameters as defined in the accelerator community. These definitions are the horizontal axis x (for the vertical axis y similar identities):

$$\begin{aligned} \beta_x &= \frac{< x^2 >}{\epsilon_x} \\ \gamma_x &= \frac{< x'^2 >}{\epsilon_x} \\ \alpha_x &= -\frac{< xx' >}{e_x} \\ \gamma_y \beta_y - \alpha_y^2 &= 1 \end{aligned} \tag{4}$$

Where ϵ_x is the geomatric emittance which can be derived from the normalized emittance $\epsilon_{x,n} = \beta_z \cdot \gamma \cdot \epsilon_x$. The squared RMS beam size is $< x^2 >$, and the squared RMS beam angular divergance is $< x'^2 >$.
Our strategy for deriving the parameters is as follows:
For the vertical axis y, the electron beam must enter the Undulator at a waist therefore the Twiss parameters are:

$$\begin{aligned} \alpha_y &= 0 \\ \beta_y &= \frac{\gamma \lambda_u}{K 2\pi} \\ \gamma_y &= \frac{1}{\beta_y} \end{aligned} \tag{5}$$

Where the K is the undulator parameter and λ_u is undulator wavelength and β_y is taken from [5].
Calculating the Twiss parameters through a drift is as follows [6]:

$$\begin{pmatrix} \beta(L) \\ \alpha(L) \\ \gamma(L) \end{pmatrix} \begin{pmatrix} 1 & 2L & L^2 \\ 0 & 1 & -L \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \alpha_0 \\ \gamma_0 \end{pmatrix} \tag{6}$$

For the horizontal axis, we assume the waist is at the center of the Undulator. We then move forward to calculating the parameters at the beginnging of the undulator from:

$$\begin{aligned} \alpha_x(L/2) = 0 = \alpha_x(0) - L/2 * \gamma_x(0) \\ \sigma_x \sim \sigma_r = \sqrt{\epsilon_r Z_r} \end{aligned} \tag{7}$$

The RMS beam size of the electron should roughly match the sigma of the radiation [4]. Where $\epsilon_r = \frac{\lambda_r}{4\pi}$ is the optical emittance, and Z_r is the Rayleigh range of the optical beam.

FEL selection of parameters:

$$\lambda_r = \frac{\lambda_u}{2\gamma^2}(1 + K^2/2) \tag{8}$$

The fundamental wavelength of the undulator radiation, γ is electron beam energy. $K = \frac{eB_0}{mck_u} = 0.934\lambda_u[cm]B_0[T]$ is undulator paramter, B_0 is undulator peak magnetic field on axis λ_u is undulator wavelength.

0.4 Desynchronization Studies

We expect laser lethargy in the oscillator FEL. When synchronization exists between the optical pulse and electron beam in the oscillator, eventually the optical pulse will lag behind the e-beam and will gain less power. In order to fight this phenomenon, the length of cavity is shortened by the insertion of the second mirror.

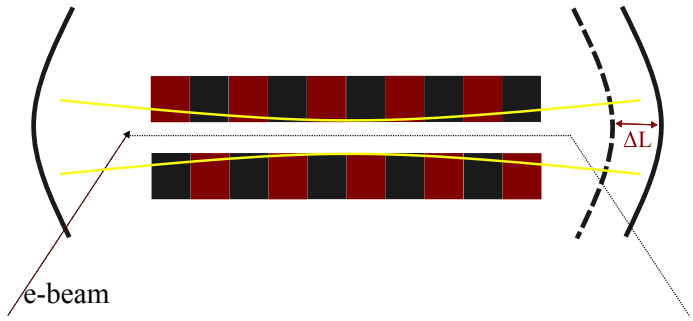

Figure 2: Cavity schematics

The desynchronization value d is defined as the ratio of the cavity shortening Fig. 2 to the slippage length [2], $d = \frac{\Delta L}{N_u \lambda_s}$, N_u is number of undulator periods. λ_s is radiation wavelength
Desynchornization is done in the code by interpolating the radiation field. The Matlab "interp1" function is used with a new time axis that is shifted by the cavity shortening length.
In the desynchronization function the square root of the cavity loss is multiplied by the field.
The desynchronization is also applied in the "$inser_lens2.m$" function by using a shorter cavity length for the field propagation.

The following are few figures generated for exemplary purposes. Fig. 3 shows the peak power of the optical pulse versus the number of passes in the oscillator. It is clearly shown that from oscillator pass 100-150 the optical pulse reaches saturation. This is shown for three different oscillator runs, each with different desynchronization value: d-0.007,d- 0.025, d-0.05. Fig. 4 displays the Peak position of the optical pulse versus the pass number in the FEL oscillator. As in the previous figure this figure also shows three different simulations, each with different desynchronization value. Fig. 5 shows the FWHM (Full Width Half Maximum) in pico seconds of the optical pulse at the end of the Undulator, for each pass through the Oscillator. As in the previous two figures, fig. 5 also displays three simulations, each simulation have different desynchronization value.

Which script generates which figure Input files and settings used for each Explanation of axes, units, transformations (e.g., theta to t, weighting by current profile for phase space plotting) Short explanation of each plot's meaning and what it shows Any known issues or open questions Script for making phase space movies, and links to movies

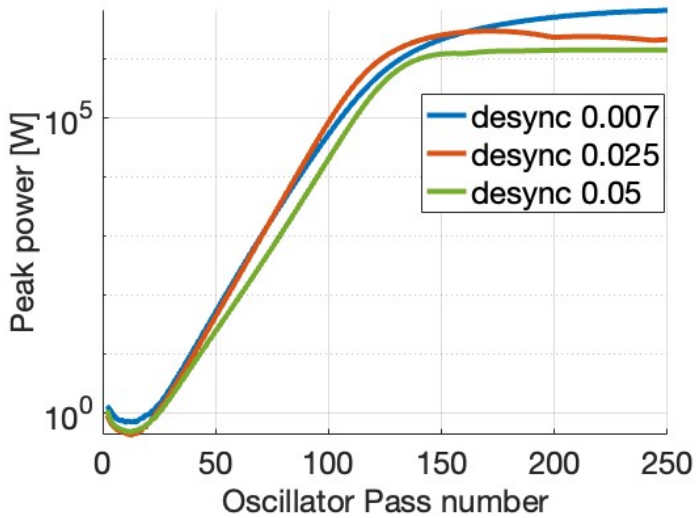

Figure 3: Optical pulse peak power vs. pass number for different desynchronization values

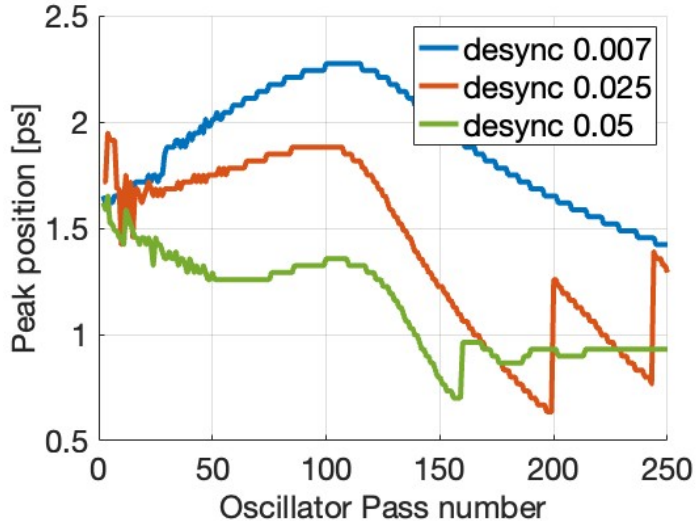

Figure 4: Peak position vs. oscillator pass number for different desynchronization values

0.5 Data Management and File Locations

The FLD, RST, and output hdf5 files are stored in the main directory, during runtime. Notice that these files are saved only every 20 oscillator pass, if required

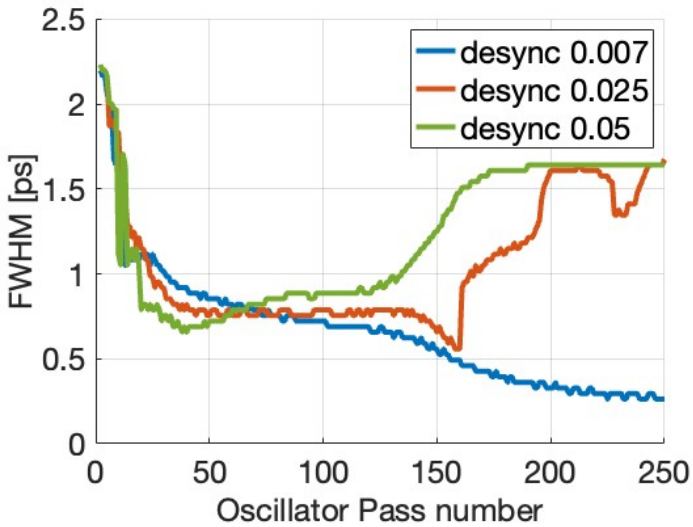

Figure 5: FWHM vs. oscillator pass number for different desynchronization values

otherwise, (for example the generation of a movie of the phase space out of RST files), a few lines in the *input_EE.m* files should be commented out:
lines 42: "if (mod(reflect,20))" 43: "reflect=2;" 44:"end"
The data for display from the add-on is saved under the "sav" directory (unless otherwise specified by the user).

0.6 Version Control and Reproducibility

The final version of the code is stored on GitHub in the following location: `github.com/amirwein/oscillator_addon`
It is possible to run the Oscillator add-on after cloning the code to your computer. Installation of `GINGER-3D` is necessary.
Some initial plots can be generated in test mode without the `GINGER-3D` package.

0.7 Future Work or Known Limitations

We suggest several plans for future work.

1. It should be possible to run the code with the GENESIS FEL ver 4 package. Code development for that purposes has already started.

2. It is also possible to create a GUI for running the code.

3. Running the code on a cluster computer.

Ideas for new simulations are to incorporate the add-on with the Michelson Interferometer configuration. A code for the interferometer for an ideal Gaussian pulse was designed and checked. The Interferometer code can be integrated with the oscillator add-on code. More effects such as phase coherence, maximal power extraction and cavity length adjustment, can be verified.

Saving the optical pulse profile in hdf5 file
Incomplete areas (e.g., plots not verified) Bugs or unexplained behavior

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